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Experiment 2: Case Study

Problem Statement: Choose a real-world case study such as an Online Food Delivery System, create user stories, select an appropriate software development process model, identify a suitable programming language, and propose an implementation and evaluation strategy.

Real-World case study: an online food delivery system

1. Functional Requirements

Customer

- Register and login securely
- Browse nearby restaurants
- Search food items by name or cuisine
- View restaurant menus and item details
- Add items to cart
- Place food orders
- Make secure online payments
- Track order status in real time
- View order history
- Rate and review restaurants
- Apply discount coupons
- Cancel an order

Restaurant Partner

- Login securely
- Manage menu items and pricing
- Update item availability
- Accept or reject incoming orders
- Update order preparation status
- View order history
- Manage restaurant profile and timings
- View sales reports

Delivery Agent

- Login securely
- View assigned delivery orders
- Update delivery status
- Navigate to pickup and delivery locations
- Mark order as delivered
- View earnings and delivery history

Administrator

- Login securely
- Manage customer accounts

- Manage restaurant partner accounts
- Manage delivery agent accounts
- Monitor orders across the platform
- Manage payment settlements
- Generate sales and performance reports
- Manage promotional offers and coupons
- Handle customer complaints and support tickets
- Backup and restore system data

2. User Stories

User Story ID	User Story	Priority
US-01	As a customer, I want to register and log in securely so that I can access the food delivery app safely.	High
US-02	As a customer, I want to browse nearby restaurants so that I can choose where to order food from.	High
US-03	As a customer, I want to search for specific food items or cuisines so that I can quickly find what I want to eat.	High
US-04	As a customer, I want to add items to my cart and place an order so that I can get food delivered to my location.	High
US-05	As a customer, I want to make secure online payments so that I can complete my order conveniently.	High
US-06	As a customer, I want to track my order in real time so that I know when my food will arrive.	High
US-07	As a customer, I want to rate and review restaurants so that I can share my experience with others.	Medium
US-08	As a restaurant partner, I want to manage my menu and item availability so that customers see accurate information.	High
US-09	As a restaurant partner, I want to accept or reject incoming orders so that I can manage my kitchen workload.	High
US-10	As a delivery agent, I want to view assigned deliveries and update their status so that customers stay informed.	High
US-11	As an administrator, I want to manage restaurant and delivery agent accounts so that only verified partners use the platform.	Medium
US-12	As an administrator, I want to generate sales and performance reports so that management can monitor platform performance.	Medium

3. Selected Software Development Process Model

Agile Scrum

Reason for Selection

- Requirements change frequently based on customer and market feedback.
- Continuous feedback from customers, restaurant partners, and delivery agents.
- Incremental delivery of platform modules.
- Faster development and testing cycles.
- Easy maintenance and bug fixing.
- Suitable for large-scale, high-traffic consumer applications.

Sprint Plan

Sprint	Features
Sprint 1	User Authentication & Registration
Sprint 2	Restaurant Listing & Menu Management
Sprint 3	Cart & Order Placement
Sprint 4	Payment Gateway Integration
Sprint 5	Order Tracking & Delivery Management
Sprint 6	Ratings, Reviews & Notifications
Sprint 7	Testing & Deployment

4. Programming Language Selection

Backend

- Java (Spring Boot)

Reasons

- Secure
- Scalable
- Enterprise-level applications
- REST API support
- Excellent database connectivity

Frontend

- HTML5
- CSS3
- JavaScript
- React.js

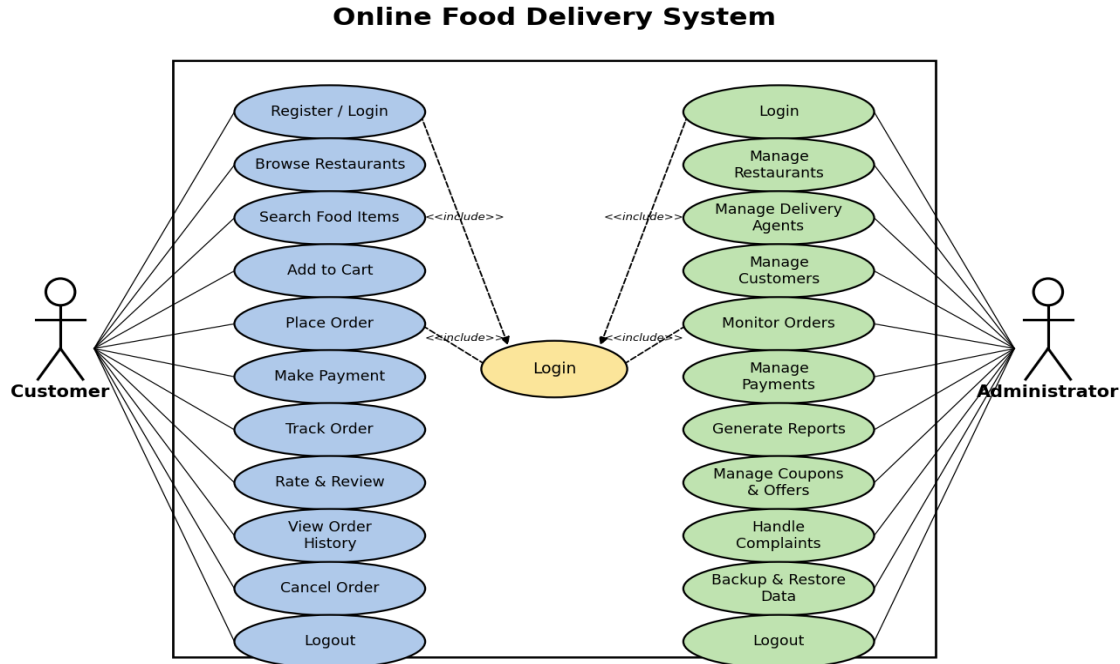
Database

- MySQL

Other Technologies

- REST API
- JWT Authentication
- Stripe/Razorpay Payment Gateway
- Google Maps API for live order tracking
- Firebase Cloud Messaging for push notifications
- Git
- Docker
- AWS Cloud

5. System Architecture



6. Implementation Strategy

Phase 1 -- Requirement Analysis

- Gather requirements from customers, restaurant partners, and delivery agents.
- Prepare the Software Requirement Specification (SRS).
- Develop use case diagrams.
- Identify system modules.

Phase 2 -- System Design

- Design database schema.
- Design user interface wireframes.
- Create UML diagrams.
- Develop overall system architecture.

Phase 3 -- Development

- Implement secure login and authentication.
- Develop Customer App / Restaurant Browsing Module.
- Develop Cart and Order Placement Module.
- Develop Payment Gateway Integration.
- Develop Delivery Tracking Module.
- Develop Ratings, Reviews and Notification Module.

Phase 4 -- Testing

Unit Testing

- Login
- Cart and Checkout
- Payment Processing
- Order Tracking

Integration Testing

- Order + Payment
- Restaurant + Order Management
- Delivery Agent + Notification System

System Testing

- Complete platform functionality

User Acceptance Testing (UAT)

- Verify all customer, restaurant, and delivery workflow requirements.

Phase 5 -- Deployment

- Docker Container
- AWS EC2
- MySQL Database
- Nginx Web Server

7. Evaluation Strategy

Evaluation Parameter	Method
Functional Accuracy	Requirement verification
Performance	Load testing using JMeter
Security	Authentication and penetration testing
Reliability	Continuous monitoring
Usability	User feedback survey
Scalability	Concurrent user / order-load testing
Maintainability	Code review and documentation

8. Expected Outcomes

- Secure and user-friendly online food ordering platform.
- Fast restaurant and food search.
- Reliable order placement and management.
- Real-time delivery tracking.
- Secure online payments.
- Improved customer satisfaction.
- Scalable architecture to handle high order volumes.